

Generation of AC

1. Which, among the following, is the correct expression for alternating emf generated?

- a) $e=2Blv\sin(\theta)$
- b) $e=2B2lv\sin(\theta)$
- c) $e=Blv\sin(\theta)$
- d) $e=4Blv\sin(\theta)$

Answer: c

Explanation: The correct expression for alternating emf generated is $e=Blv\sin(\theta)$. Where B stands for magnetic field density, l is the length of each of the parallel sides v is the velocity with which the conductor is moved and θ is the angle between the velocity and the length.

Calculate the maximum emf when the velocity is 10m/s, the length is 3m and the magnetic field density is 5T.

- a) 150V
- b) 100V
- c) 300V
- d) 0V

Answer a

What should theta be in order to get maximum emf in sinusoidal voltage?

- a) 0
- b) 90
- c) 180
- d) 45

Answer b

In an A.C. generator, increase in number of turns in the coil _____

- a) Increases emf
- b) Decreases emf
- c) Makes the emf zero
- d) Maintains the emf at a constant value

AC values

The r.m.s. value of alternating current is given by steady (D.C.) current which when flowing through a given circuit for given time produces

- (a) the more heat than produced by A.C. when flowing through the same circuit
- (b) the same heat as produced by A.C. when flowing through the same circuit
- (c) the less heat than produced by A.C. flowing through the same circuit
- (d) none of the above

Ans: b

The power consumed in a circuit element will be least when the phase difference between the current and voltage is

- (a) 180°
- (b) 90°
- (c) 60°
- (d) 0°

Ans: b

The r.m.s. value and mean value is the same in the case of

- (a) triangular wave
- (b) sine wave
- (c) square wave
- (d) half wave rectified sine wave

Ans: c

For the same peak value which of the following wave will have the highest r.m.s. value ?

- (a) square wave
- (b) half wave rectified sine wave
- (c) triangular wave
- (d) sine wave

Ans: a

For the same peak value, which of the following wave has the least mean value ?

- (a) half wave rectified sine wave
- (b) triangular wave
- (c) sine wave
- (d) square wave

Ans: a

The peak value of a sine wave is 200 V. Its average value is

- (a) 127.4 V
- (b) 141.4 V
- (c) 282.8 V
- (d) 200V

Ans: a

The voltage of domestic supply is 220V. This figure represents

- (a) mean value
- (b) r.m.s. value
- (c) peak value
- (d) average value

Ans: a

The r.m.s. value of a sinusoidal A.C. current is equal to its value at an angle of _____degrees.

- (a) 90
- (b) 60
- (c) 45
- (d) 30

Ans: c

The ratio of effective value to average value is called the factor.

- form
- peak
- average
- Q-factor

Answer: form

The RMS value of a sine wave is 100 A. Its peak value is

- 70.7 A
- 141 A
- 150 A
- 282.8 A

Ans 141A

R/L/C circuit

In a series RLC circuit, the phase difference between the current in the capacitor and the current in the resistor is?

- a) 0°
- b) 90°
- c) 180°
- d) 360°

Answer: a

Explanation: In a series RLC circuit, the phase difference between the current in the capacitor and the current in the resistor is 0° because same current flows in the capacitor as well as the resistor.

In a series RLC circuit, the phase difference between the current in the circuit and the voltage across the capacitor is?

- a) 0°
- b) 90°
- c) 180°
- d) 360°

Answer: b

Explanation: In a series RLC circuit, voltage across capacitor lags the current in the circuit by 90° so, the phase difference between the voltage across the capacitor and the current in the circuit is 90°

Which, among the following is the correct expression for impedance?

- a) $Z=Y$
- b) $Z=1/Y$
- c) $Z=Y/2$
- d) $Z=1/2Y$

Answer: b Explanation: We know that impedance is the reciprocal of admittance, hence the correct expression for impedance is: $Z=1/Y$

In a pure resistive circuit

- A. Current lags behind the voltage by 90°
- B. Current leads the voltage by 90°
- C. Current can lead or lag the voltage by 90°
- D. Current is in phase with the voltage

Answer:- D. Current is in phase with the voltage

For a purely resistive circuit the following statement is in correct

- A. Work done is zero

- B. Power consumed is zero
- C. Heat produced is zero
- D. Power factor is unity

Answer:- D. Power factor is unity

A $90\ \Omega$ resistor, a coil with $30\ \Omega$ of reactance, and a capacitor with $50\ \Omega$ of reactance are in series across a 12 V ac source. The current through the resistor is

- a) 9 mA
- b) 90 mA
- c) 13 mA
- d) 130 mA

Answer 130mA

How many types of power can be defined in an AC circuit?

- a) 3
- b) 2
- c) 1
- d) 5

Answer 3

Which of the following is true about power factor?

- a) $\sin\Phi = \text{Truepower} / \text{Apparentpower}$
- b) $\cos\Phi = \text{Truepower} / \text{Apparentpower}$
- c) $\sin\Phi = \text{Apparentpower} / \text{Truepower}$
- d) $\cos\Phi = \text{Apparentpower} / \text{Truepower}$

Answer: b

What is the power factor in a pure inductive or capacitive circuit?

- a) -1
- b) 0
- c) 1
- d) Infinity

Answer b

Unit of Power factor is

- a) watt
- b) unit less
- c) Ampere
- d) VAR

Answer b

Active power and apparent power are respectively represented by?

- a. kW and kVAR
- b. kVAR and kVA
- c. kVA and kVAR
- d. kW and kVA

Answer d

Three phase system

Electric power in a Three Phase Circuit = _____.

- a) $P = 3 V_{Ph} I_{Ph} \cos\Phi$
- b) $P = \sqrt{3} V_L I_L \cos\Phi$
- c) Both 1 & 2.
- d) None of The Above

Answer C

In a three phase AC circuit, the sum of all three generated voltages is _____ ?

- a) Infinite (∞)
- b) Zero (0)
- c) One (1)
- d) None of the above

Answer b

For a star connected three phase AC circuit _____ ?

- a) Phase voltage is equal to line voltage and phase current is three times the line current
- b) Phase voltage is square root three times less than line voltage and phase current is equal to line current
- c) Phase voltage is equal to line voltage and line current is equal to phase current
- d) None of the above

Answer b

In a three phase, delta connection _____ ?

- a) line current is equal to phase current
- b) Line voltage is equal to phase voltage
- c) None of the above
- d) Line voltage and line current is zero

Answer b
